On Model Design of a Surrogate Fuel Formulation

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Calculations of evaporation characteristics (distillation curve, two-phase diagram, and critical points) of surrogates are described in detail. The efficiency of some surrogate blends, represented in literature, in reflecting the evaporation characteristics was analyzed. Based on the analysis, the chemical capabilities of surrogate models are not linked to their abilities to reflect the phase-equilibrium properties of real fuel. It is shown that model design of practical fuels must include the phase-equilibrium and distillation curve calculations. A surrogate mixture was selected, which closely matches the boiling-point curve and two-phase diagram for jet-A. Next, physical properties of reference fuel were taken into consideration: combustion enthalpy, formation enthalpy, molar weight, approximate formula (carbon per hydrogen ratio), sooting tendency index, critical point, two-phase diagram, and distillation curve. [DOI: 10.1115/1.4000593]

1 Introduction

Practical fuels, such as gasoline, kerosene, diesel, and so on, are complex mixtures of several hundred individual species, the majority of which are distilled from crude oil. Therefore, only simplified chemical kinetic models of practical fuels (surrogate blends or reference fuels) can be used in the design of engine combustion chambers. Standard rules for the construction of an input content formula of a surrogate mixture are not yet developed, but the general principles of such a construction are evident and require the reference fuel to have the essential physical and chemical properties of the real fuel. Here, under the physical properties we understand density, fuel energy content, evaporation characteristics (boiling range, flash point, and vapor pressure), thermal conductivity, surface tension, viscosity, and so on. The chemical properties relate to the chemical composition, carbon per hydrogen (C/H) ratio, flow reactor concentration histories, flame speeds, ignition delays, sooting tendency, and so on. The components of the surrogate fuel and their amounts or percentages (input formula of a surrogate fuel model) are to be specified based on the list of chemical and physical properties that need to be predicted. The number of components in the input formula of the surrogate has to be kept reasonably small (usually three to ten). Generally, at this stage of the model development we must do the first step in the model reduction and include in the surrogate mixture whenever possible only the species that are essential for the representation of physical and chemical properties of the modeled fuels.

Since one of the main aims of the combustion modeling is the simulation of fuel heat release (flame speed and ignition delay times) and pollution formation, it is easy to determine the "chemical properties" required for the construction of the input surrogate formula. The "physical properties," most closely connected to the chemical properties, are density and fuel enthalpy. These two quantities are additive and do not influence remarkably the number of hydrocarbons in the reference fuel, as the density and enthalpy values of individual hydrocarbons involved in the practical fuel combustion are quite similar [1].

Other physical characteristics, such as the evaporation and viscosity, have a greater impact on the number of components in the surrogate blend and are at the same time difficult to predict. They are very important for the precombustion processes in the chambers with short residence times of the propellants and high efficiency requirements such as liquid propellant rocket engines or jet engines. Due to the influence of surface tension and density gradients on the fuel atomization and mixing, all other physical properties are more or less closely linked to these two. A large number of surrogate fuel mixtures have been proposed recently, evaluated in engines, in fundamental experiments, and studied numerically. Most of those were developed for various jet fuels [1–12]. These models simulate the fuel oxidation chemistry and generally, successfully reflect the combustion kinetic. Unfortunately, the capabilities of the surrogate models to reflect the physical properties, especially the evaporation characteristic, are considerably less studied. The present work is aimed to elaborate criteria for the determination of the input formula of the reference fuel model. It is our first step to establish the principles of model design of surrogate blends.

This work is concentrated on the evaluation of evaporation characteristics of reference fuel models. In the first part of this paper, we briefly describe the boiling characteristics of hydrocarbons used in the surrogate blends. The second part is devoted to the methods employed for the calculation of phase diagrams, critical points, and distillation curves of multicomponent mixtures. The third part contains the thermodynamic analysis of some surrogate blends presented in literature and the evaluated composition of a sample surrogate blend, which reflects the evaporation properties of the aviation fuel.

2 Reference Fuel Evaporation Characteristics

The hydrocarbons of practical fuels are primarily paraffins, naphthenes (cycloparaffins), and aromatics. In Table 1 we summarize the hydrocarbons used in different chemical models for gasoline, kerosene, and diesel [1–12]. These species successfully represent hundreds of hydrocarbons in surrogate mixtures by simulations of fuel oxidation. How do these species represent the evaporation characteristics? What kind of criteria do we have to evaluate these characteristics?

First, the requirements for the fuel volatility are traditionally preset by the distillation curve (temperature boiling-point (TBP) curve—boiling temperature versus the volume of evaporated fraction during distillation), which represents the distribution of species by boiling temperatures. Quite often, fuel is characterized by the maximum evaporated percent volume only for some boiling temperatures instead of the distillation curve. The most important distillation characteristics are the temperatures at the top end of the range, as these provide a measure of the proportion of heavier components, which correlate closely with aromatic contents in particular. Heavier components in the fuel have more potential for incomplete vaporization and combustion, which results in the increased production of smoke or soot. Presetting lower end values

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Table 1 Hydrocarbons used in surrogate fuel models

Primary reference fuel components:	Other candidates:
Gasoline [4,8]	
Toluene	Ethanol
n-heptane	Iso-butane
Iso-octane	n-pentane
	Iso-pentane
	Cyclopentane
	Cyclohexane
	m-xylene
	Methyl-cyclohexane
Jet [2,6–12]	
Benzene	Tetralin
Toluene	Butylbenzene
Methylcyclohexane (MCH)	Tetramethylbenzene
n-heptane	Decalin
Ethyl benzene	n-decane
m-xylene	Methyl-naphthalene
Iso-octane	n-undecane
Cyclooctane	n-dodecane
Propylbenzene	n-octyl-cyclopentane
1,2,4-trimethylbenzene	n-tetradecane
n-propylcyclohexane	n-pentadecane
n-nonane	n-hexadecane
Naphthalene	
Diesel [3]	
Heptamethylnonane (HMN)	Ethyl-/
n-hexadecane	Propyl-/
	Butylcyclohexane
	Ethyl-/
	Propyl-/
	Butyl benzene
	Toluene
	Methyl-cyclohexane
	n-heptane
	Iso-octane
	Decalin
	Tetralin
	Methyl-naphthalene
	n-decane
	n-dodecane
	Tetradecane
	n-decylbenzene
	Iso-cetane

Table 2 Thermodynamical properties of hydrocarbons used in surrogate fuel models [13]

Formula	Name	T_{boil}	T_{crit}	P_{crit}
	n- and iso-paraffins			
$CH(CH_3)_3$	Iso-butane	261.34	407.85	36.4
C_5H_{12}	n-pentane	309.22	469.7	33.7
C_5H_{12}	Iso-pentane	300.99	460.39	33.81
C_6H_{14}	Hexane	341.88	507.6	30.25
C_7H_{16}	Heptane	371.57	540.2	27.4
C_7H_{16}	2-methylhexane	363.18	530.1	27.3
C_8H_{18}	3et2metpentane	388.81	567	27
C_8H_{18}	Iso-octane	372.39	543.9	25.7
C_9H_{20}	Nonane	423.97	594.6	22.9
$C_{10}H_{22}$	n-decane	447.3	617.7	21.1
$C_{11}H_{24}$	Undecane	469.08	639	19.8
$C_{12}H_{26}$	Dodecane	489.48	658	18.2
$C_{13}H_{28}$	Tridecane	508.63	675	16.8
$C_{14}H_{30}$	Tetradecane	526.76	693	15.7
$C_{15}H_{32}$	Pentadecane	543.83	708	14.8
$C_{16}H_{34}$	Hexadecane	559.98	723	14
$C_{16}H_{34}$	HMN	520.25	693	15.7
	Cycloparaffins			
C_6H_{12}	Cyclohexane	353.93	553.5	40.73
C_7H_{14}	Methylcyclohexane	374.09	572.19	34.71
C_8H_{10}	M-xylene	412.34	617	35.41
C_8H_{16}	Cyclooctane	424.31	647.2	35.7
C_8H_{16}	Ethylcyclohexane	405	609	
C_9H_{18}	Propylcyclohexane	429.9	639	27.7
$C_{10}H_{18}$	Trans-decalin	460.42	687	46
	Aromatics			
C_6H_6	Benzene	353.24	562.05	48.95
C_7H_8	Toluene	383.79	591.75	41.08
C_8H_{10}	Ethylbenzene	409.36	617.15	36.09
C_9H_{12}	Propylbenzene	432.35	638.35	32
C_9H_{12}	1,2,4-trimethylbenzene	442.5	649.1	32.32
$C_{10}H_{12}$	Tetralin	480.75	720	36.5
$C_{11}H_{10}$	1-methyl-naphthalene	517.84	772	36

of the distillation curve, which correspond to higher temperatures, reduces the proportion of the heavy components and produces a cleaner burning. It can also reduce the density and the viscosity of the fuel as these properties are closely connected. If we want to describe the distillation curve for a mixture, we have to select a set of hydrocarbons with the boiling temperatures lying between the lowest and highest temperatures of the distillation curve. The boiling temperatures of individual hydrocarbons used in surrogate fuel models are provided in Table 2. Evidently, one can generate a variety of different sets, which will describe the given boiling range.

The distillation curve for an individual fuel is usually obtained from experiments [14–17] and these data can be used for the validation of the surrogate blend content [10,17].

Examples of TBP curves are presented in Figs. 1–3 revealing that different fuel grades have different distillation curves. We would like to note that there are no sufficient data for distillation curves and that it is difficult to evaluate experimental errors in available measurements or to find data related to the selected fuel.

Second, the phase behavior (e.g., vapor-liquid equilibrium and accordingly surface tension and viscosity) over the full ranges of temperature and pressure is characterized by two-phase diagrams, Fig. 4. Unfortunately, the situation with experimental two-phase diagrams is even worse than that with distillation curves—we did not find other measurements of liquid-gas equilibrium for real fuel besides data reported in Ref. [8], and for gasoline we did not find those at all. The phase equilibrium defines more the chemical

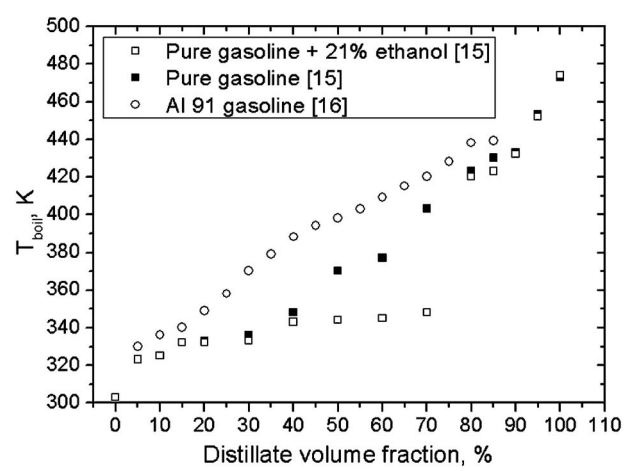

Fig. 1 Distillation curves for gasoline samples: pure gasoline, gasoline with 21% added ethanol from Ref. [15] and distillation curves of the 91 AI [16]

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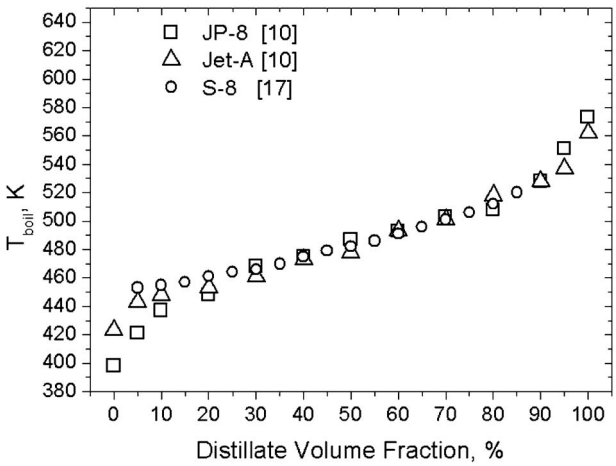

Fig. 2 Distillation curves for kerosene samples: experimental distillation curve for S-8 at 83 kPa, adjusted for volume shift from Ref. [17] and boiling-point curves for jet-A and JP-8 fuels from Ref. [10]

composition of a surrogate mixture: blends and real fuel must have identical boiling range/phase behavior and critical temperature and pressure.

These parameters are not additive and are not easy to fit. They are calculated from the equation of state (EOS), i.e., are linked directly to the thermodynamic state functions. Ideally, for well-controlled modeling, we have to approximate the thermodynamic functions of a multicomponent real fuel to those of the simplified surrogate blend. In what follows, we made an attempt to identify tools to calculate thermodynamic equilibrium, i.e., two-phase diagrams and critical points, for multicomponent mixtures. Using these tools we calculated the two-phase diagrams for some surrogates from literature and optimized the chemical contents of a blend, which satisfies the thermodynamic properties of practical fuels.

Calculating vapor-liquid equilibrium for mixtures, one faces two main problems. The first one is choosing an equation of state, which can provide reliable results and at the same time can be, possibly, simple and would not require additional calculations to determine the equation parameters. The second problem is choosing a mixing rule to evaluate EOS parameters for multicomponent mixtures. For nonideal cases (a big difference in molecular size, high polarity, and so on) simple mixing rules are not applicable.

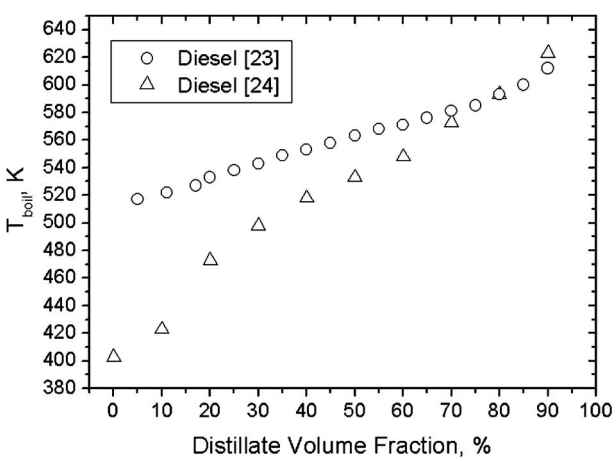

Fig. 3 Distillation curves for diesel samples: experimental distillation curves from Refs. [23,24]

3 Choosing of an Equation of State

At present there are dozens of different equations of state, which can be divided into four groups beginning from simple ones (Peng–Robinson and Redlich–Kwong) to more complex (Patel, Zabaloy, and Vera) perturbation models. A detailed review and comparison of equations of state and mixing rules can be seen in Refs. [13,18–22], Table 3.

3.1 Virial Equation of State. The virial equation of state has a theoretical basis, its coefficients relate to intermolecular forces

$$Z = 1 + \frac{B}{v} + \frac{C}{v^2} + \frac{D}{v^3} \cdots \tag{1}$$

where $B, C, D\ldots$ are the second, third, fourth… virial coefficients.

3.2 Cubic Equation of State. The general form is

$$P = \frac{\mathrm{R}T}{v - b} - \frac{a}{v^2 + \gamma v + \varepsilon} \tag{2}$$

where $a, b, \gamma, \varepsilon$ are the parameters.

Various modifications of cubic equations (van der Waals, Soave–Redlich–Kwong (SRK), and Peng–Robinson) are widely applied because all equation parameters can be easily evaluated from critical temperatures and pressures of studied substances. Martin [20] and Trebble and Bishnoi [21] carried out a detailed comparison of the different cubic equations of state.

3.3 Benedict–Webb–Rubin and Modified Benedict–Webb–Rubin Equations of State. The general form is

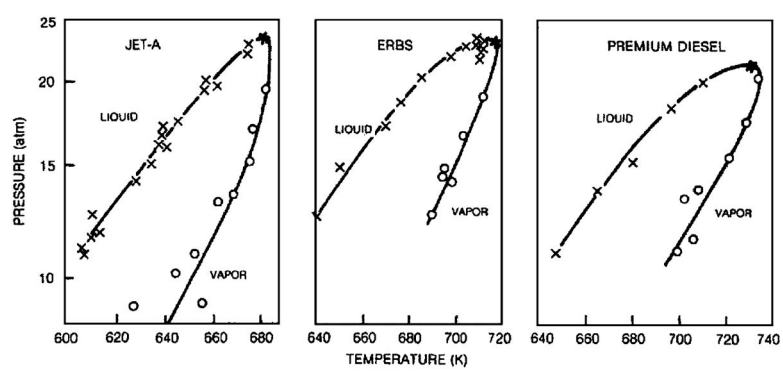

Fig. 4 Phase diagram for jet-A and diesel fuels from Ref. [8]; ∗ symbols used for critical points, ○ symbols used for bubble points, and × symbols used for dew points

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Table 3 Comparison of the main types of equations of state [13,18–22] (modified Benedict–Webb–Rubin (MBWR))

EOS	Advantage	Disadvantage
Virial equation of state	• Relatively simple	• Not precise at high pressure • Coefficients are very difficult to determine • Usually can be applied only for a single-phase gas system
Cubic	• Simple • Analytic expressions for density or fugacity are not too demanding	• Adequate, but does not provide perfect accuracy • For polar and associating substances, four and more parameters should be used • Inapplicable for mixtures of fluids with big differences in molecular size
BWR and MBWR	• Good precision	• Many parameters should be calculated • Results depend strongly on errors in experimental data used for the parameter evaluation • Generalized modifications (with a small amount of parameters) are not accurate
Based on the perturbation theory (perturbed hard chain, statistical associated fluid, and perturbed hard chain of spheres)	• Good precision • Applicable for mixtures of substances with big differences in molecular size • Applicable for high pressures	• Very complicated • For mixtures of unlike molecules, three or four pure-component parameters are needed

$$Z = 1 + f_1(T)/V + f_2(T)/V^2 + f_3(T)/V^n + f_4(T) \times[(\gamma_1 + \gamma_2/V^2)/V^m]\exp(-\gamma_2/V^2) \quad (3)$$

where functions f_i can contain up to 30 parameters and γ_1, γ_2 are additional parameters. It provides a high level of equation flexibility and accuracy. This resulted in the development of many modifications, which are widely used for engineering purposes. The Span–Wagner modification [25]

$$\frac{p}{\rho \mathrm{R}T} = \sum_{i=1}^{I_{\mathrm{pol}}} n_i \tau^{t_i} \xi^{d_i} + \sum_{i=I_{\mathrm{pol}}+1}^{I_{\mathrm{pol}}+I_{\mathrm{exp}}} n_i \tau^{t_i} \xi^{d_i} \exp(-\gamma \xi^2) \quad (4)$$

where values of I_{pol}, I_{exp}, t_i, and d_i are defined in the "trial and error" procedure.

3.4 Perturbation Models. Compressibility factor $\boldsymbol{Z}$ is found using the thermodynamic partial derivation of residual Helmholtz energy [13]

$$Z = 1 - v \left. \frac{\partial(A^r/\mathrm{R}T)}{\partial v} \right|_{\tau} \quad (5)$$

Molecules are considered as a collection of functional groups. Residual Helmholtz energy is written as

$$A^r/\mathrm{R}T = [A_R^r(T,v)/\mathrm{R}T] + \sum [A_{\mathrm{att}}^r(T,v)/\mathrm{R}T]^{(i)} \quad (6)$$

where perturbation terms $[A_{\mathrm{att}}^r(T,v)/\mathrm{R}T]$ can be obtained from a rigorous or precise theory. For example, in the statistical associated-fluid theory, residual Helmholtz energy consists of Helmholtz energies of the ideal gas A_{ideal}, the molecule chains A_{chain}, the molecule segments A_{segm}, and the associations A_{assoc} [18,26]

$$A^r = A_{\mathrm{ideal}} + A_{\mathrm{segm}} + A_{\mathrm{chain}} + A_{\mathrm{assoc}} \quad (7)$$

4 Choosing Mixing Rules

There is a multitude of mixing rules for the EOS parameters proposed: beginning from simple van der Waals rules to complicated relations containing Gibbs energy.

van der Waals mixing rule for an arbitrary parameter Ψ

$$\Psi_m = \sum_{i=1}^{n} \sum_{j=1}^{n} y_i y_j \Psi_{ij} \quad (8)$$

where cross parameter Ψ_{ij} is defined by combining rules

$$\Psi_{ij} = \frac{\Psi_{ii} + \Psi_{jj}}{2}(1 - l_{ij}) \quad (9)$$

if $\Psi = b$ from Eq. (2) and

$$\Psi_{ij} = \sqrt{\Psi_{ii}\Psi_{jj}}(1 - k_{ij}) \quad (10)$$

if $\Psi = a$ from Eq. (2). l_{ij} and k_{ij} are found by fitting into the experimental data, if they are not available they are usually considered zero. This mixing rule also has many composition-dependent modifications [27–32].

For a more precise result or in case of a strong nonideality of substances, excess Gibbs energy (G^E) mixing rules should be applied [19]. A detailed review and comparison of equations of state and mixing rules can be found in Refs. [13,18,19,26].

The difference in size of hydrocarbons used in reference blends rarely exceeds factor 2; also, hydrocarbons are nonpolar molecules with well established available thermodynamical parameters. For those reasons, we can apply the simple cubic equation of state (Soave modification of the Redlich–Kwong EOS) and van der Waals mixing rules to two-phase diagram calculations with acceptable accuracy.

5 Calculations of Reference Fuel Evaporation Characteristics

5.1 Phase Diagram. The developed method of phase diagram calculation is based on the works of Michelsen [22,33] and Asselineau et al. [34].

The used SRK equation of state is

$$P = \frac{\mathrm{R}T}{v - b} - \frac{a}{v(v + b)} \quad (11)$$

The fugacity coefficient for SRK EOS can be calculated as

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$$\ln(\varphi_i) = -\ln\left(z\left(1-\frac{b_{\text{mix}}}{\nu}\right)\right) + \frac{b_i}{b_{\text{mix}}}(z-1) + \frac{a_{\text{mix}}}{b_{\text{mix}}\text{R}T}\left(\frac{b_i}{b_{\text{mix}}} - \frac{2\sum_{j=1}^{N}\sqrt{a_i a_j}x_j}{a_{\text{mix}}}\right)\ln\left(1+\frac{b_{\text{mix}}}{\nu}\right) \tag{12}$$

van der Waals mixing rules give for the EOS parameter calculations

$$a_{\text{mix}}^L = \sum_{i=1}^{N}\sum_{j=1}^{N} x_i x_j a_{ij}, \quad b_{\text{mix}}^L = \sum_{i=1}^{N} x_i b_i$$
$$a_{\text{mix}}^V = \sum_{i=1}^{N}\sum_{j=1}^{N} y_i y_j a_{ij}, \quad b_{\text{mix}}^V = \sum_{i=1}^{N} y_i b_i \tag{13}$$

where

$$\kappa_i = 0.48508 + 1.5571\omega_i - 0.15613\omega_i^2,$$
$$\alpha_i = \left[1+\left(1-\kappa_i\sqrt{\frac{T}{T_{\text{crit},i}}}\right)\right]^2$$
$$a_i = 0.42747\alpha_i\frac{\text{R}^2T_{\text{crit},i}^2}{P_{\text{crit},i}^2}, \quad b_i = 0.08664\frac{\text{R}T_{\text{crit},i}}{P_{\text{crit},i}} \tag{14}$$
$$a_{ij} = \sqrt{a_i a_j}(1-k_{ij})$$

Parameter k_{ij} is found by minimizing the discrepancy of experimental and calculated thermodynamic data. The strategy for the two-phase equilibrium determination can be explained as follows: from a given composition of liquid phase x_i and system pressure P (bubble point T problem) we have to calculate the composition of gas phase y_i and corresponding equilibrium temperature T. We have $N+3$ (N is the number of substances) unknowns: $y_{i,i=1\ldots N}$, T, and the compressibilities of liquid and gas phases—z_L, z_V; and $N+3$ equations: two equations of state, for liquid and gas phases, and N equilibrium conditions

$$f_i^V = y_i\phi_i^V P = y_i\phi_i^L P = f_i^L, i = 1\ldots N \tag{15}$$

and one equation for y_i

$$\sum_{i=1}^{N} y_i = 1 \tag{16}$$

Now we can form a system of equations to be solved

$$F(1) = P - \frac{\text{R}T}{v-b_{\text{mix}}^L} + \frac{a_{\text{mix}}^L}{v(v+b_{\text{mix}}^L)}$$
$$F(2) = P - \frac{\text{R}T}{v-b_{\text{mix}}^V} + \frac{a_{\text{mix}}^V}{v(v+b_{\text{mix}}^L)}$$
$$F(3) = \varphi_1^L x_1 - \varphi_1^V y_1$$
$$\ldots$$
$$F(n+1) = \varphi_{n-1}^L x_{n-1} - \varphi_{n-1}^V y_{n-1}$$
$$F(n+2) = \varphi_n^L x_n - \varphi_n^V\left(1-\sum_{j=1}^{n-1} y_j\right) \tag{17}$$

for vector of variables $\boldsymbol{V}^T = (T, z^L, z^V, y_1, y_2, \ldots, y_{N-1})$.

The algorithm of the two-phase diagram construction is realized with the multidimensional Newton–Raphson method. Since this method is very sensitive to the initial solution, we must have a procedure for calculating the initial point: $(T_0, z_0^L, z_0^V, y_1^0, y_2^0, \ldots, y_{N-1}^0)$. As suggested by Mollerup (cited in Ref. [33]), K factors (ratio y_i to x_i) at low pressures are well approximated by

$$\text{K}_i = \frac{y_i}{x_i} = \frac{P_{\text{crit}_i}}{P}\exp\left[5.42\left(1-\frac{T_{\text{crit}_i}}{T}\right)\right], \tag{18}$$

Adding to these equations $\Sigma_{i=1}^N y_i = 1$, we get an equation system of $(N+1)$ equations with $(N+1)$ unknowns $(y_i, i=1\ldots N; T)$ and then using equations of state we can calculate z_L, z_V (as the smallest and the biggest roots of the corresponding EOS).

Figure 5(*a*) shows a schematic diagram for the proposed algorithm. For the dew curve the same algorithm is used, except that we have y_i and calculate x_i. Using this algorithm we were able to plot phase diagrams from low pressures to critical points for a large number of mixtures composed of up to 20 substances. Computation usually takes several seconds.

5.2 Critical Points. The calculation of critical points is based on the works of Heidemann and co-workers [35,36]. The critical point is placed on the stability limit of the stable phase. To find this stability limit, Helmholtz energy is expanded in a Taylor series around the test point

$$A - A_0 - \sum_i^N (\partial A/\partial n_i)\Delta n_i = \frac{1}{2}\sum_j^N\sum_i^N\left(\frac{\partial^2 A}{\partial n_i\,\partial n_j}\right)\Delta n_i\Delta n_j + \frac{1}{6}\sum_k^N\sum_j^N\sum_i^N\left(\frac{\partial^3 A}{\partial n_i\,\partial n_j\,\partial n_k}\right)\Delta n_i\Delta n_j\Delta n_k + O(\Delta n^4) \tag{19}$$

The stability limit of the stable phase (critical point) is mainly defined by the quadratic term in Eq. (19). If the quadratic term equals zero, the stability is defined by cubic, quaternary, and following terms, and with high accuracy, only the cubic term can be taken into consideration. Therefore, the critical point is achieved if both the quadratic and cubic terms equal zero. Helmholtz energy can be expressed through fugacity

$$\frac{\partial^2 A}{\partial n_i\,\partial n_j} = \text{R}T\frac{\partial \ln f_i}{\partial n_j} \tag{20}$$

and

$$\frac{\partial^3 A}{\partial n_i\,\partial n_j\,\partial n_k} = \text{R}T\frac{\partial^2 \ln f_i}{\partial n_j\,\partial n_k} \tag{21}$$

Therefore, the system lies on the limit of stability when

$$\det(\boldsymbol{Q}) = 0 \tag{22}$$

where matrix $\boldsymbol{Q}$ is formed by elements

$$\boldsymbol{Q}_{ij} = \left.\frac{\partial^2 A}{\partial n_i\,\partial n_j}\right|_{T,v} = \text{R}T\left.\frac{\partial \ln f_i}{\partial n_j}\right|_{T,v} \tag{23}$$

or what is the same system of equations

$$\boldsymbol{Q}\cdot\boldsymbol{\Delta n} = 0 \tag{24}$$

should have a solution $\boldsymbol{\Delta n} = (\Delta n_1, \Delta n_2, \ldots, \Delta n_N)^T$.

The definition of the critical point requires that with $\boldsymbol{\Delta n}$ found from Eq. (24) and inserted into Eq. (19), the first nonvanishing term should be of even order; therefore, the cubic term should be equal to zero

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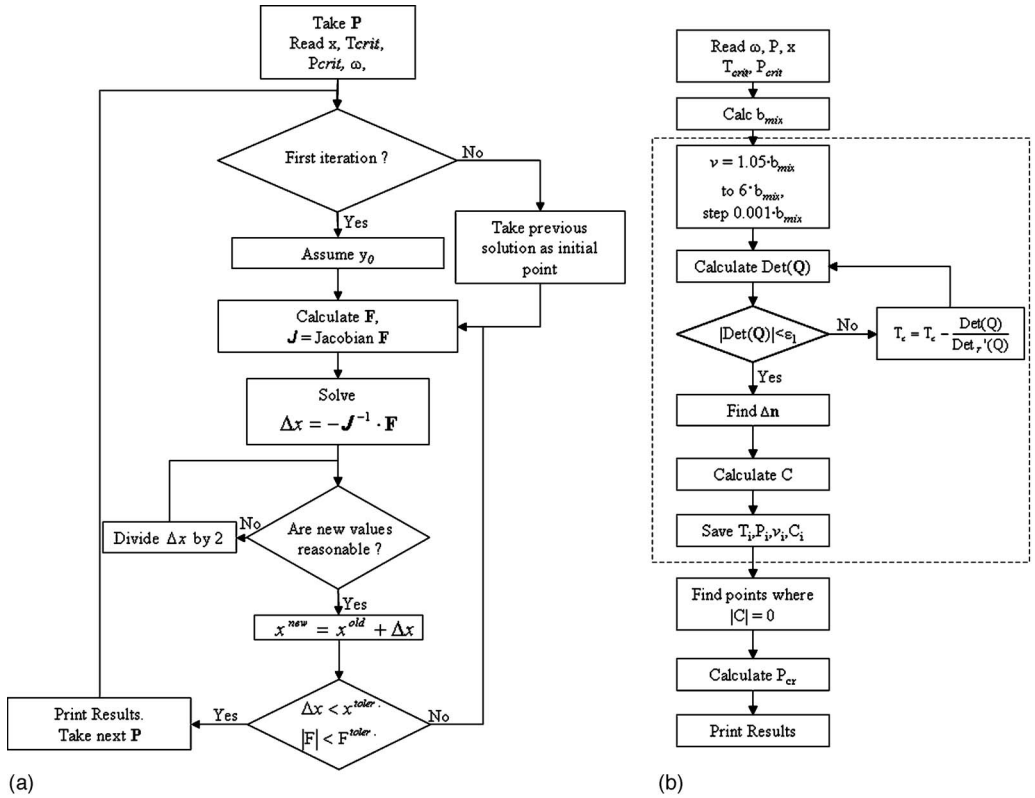

Fig. 5 Schematic diagram of proposed algorithms for (*a*) phase diagram calculation and (*b*) critical point calculation

$$C=\sum_{k}^{N}\sum_{j}^{N}\sum_{i}^{N}\left(\frac{\partial^3 A}{\partial n_i\,\partial n_j\,\partial n_k}\right)\Delta n_i\Delta n_j\Delta n_k=0 \tag{25}$$

Constants of pure components and mixing rules are similar to those used for a phase diagram construction. For the case of SRK EOS elements of matrix $\boldsymbol{Q}$ became

$$\boldsymbol{Q}_{ij}n_T=\mathrm{R}T\left(\frac{\delta_{ij}}{x_i^2}+(\beta_i+\beta_j)F_1+\beta_i\beta_jF_1^2\right)+\frac{a_{\mathrm{mix}}}{b_{\mathrm{mix}}}\left(\beta_i\beta_jF_3-\frac{a_{ij}}{a_{\mathrm{mix}}}F_5+(\beta_i\beta_j-\alpha_i\beta_j-\alpha_j\beta_i)F_6\right) \tag{26}$$

where

$$\alpha_i=\frac{1}{a_{\mathrm{mix}}}\sum x_ja_{ij},\quad \beta_i=\frac{b_i}{b_{\mathrm{mix}}} \tag{27}$$

and cubic term (25) can be calculated using the following expression:

$$Cn_T^2=\mathrm{R}T\left(\sum_{i=1}^{N}\frac{\Delta n_i^3}{x_i^2}+3\bar{N}(\bar{\beta}F_1)^2+2(\bar{\beta}F_1)^3\right)+\frac{a_{\mathrm{mix}}}{b_{\mathrm{mix}}}(3\bar{\beta}^2(2\bar{\alpha}-\bar{\beta})\times(F_3+F_6)-2\bar{\beta}^3F_4-3\bar{\beta}\bar{a}F_6) \tag{28}$$

where

$$\bar{N}=\sum_i\Delta n_i,\quad \bar{\alpha}=\sum_i\Delta n_i\alpha_i$$
$$\bar{\beta}=\sum_i\Delta n_i\beta_i,\quad \bar{a}=\frac{1}{a_{\mathrm{mix}}}\sum_i\sum_j\Delta n_i\Delta n_ja_{ij} \tag{29}$$

and

$$F_1=\frac{1}{\mathrm{K}-1},\quad F_2=\frac{2}{\mathrm{K}+1}$$
$$F_3=2\left(\frac{1}{\mathrm{K}+1}\right)^2,\quad F_4=2\left(\frac{1}{\mathrm{K}+1}\right)^3 \tag{30}$$
$$F_5=2\ln\left(\frac{\mathrm{K}+1}{\mathrm{K}}\right),\quad F_5=F_2-F_5,\quad \mathrm{K}=\frac{v}{b_{\mathrm{mix}}}$$

Figure 5(*b*) shows a schematic flowchart for the critical point calculation procedure. Such an algorithm allows for the calculation of all critical points in the area of interest.

For small mixtures (up to five substances) calculation time is several seconds, whereas for complicated mixtures (more than 20 substances) it can reach a minute.

5.3 Distillation Curve. Distillation curves can be easily predicted using the phase diagram algorithm. The equation for material balance for species i in liquid phase

$$\frac{d(Mx_i)}{dt}=-Vy_i \tag{31}$$

where M is the total number of moles in the liquid phase and V is the molar flow rate of evaporation. The first boiling point is found for the initial mixture $x_i^{(0)}$ and then the composition $x_i^{(k)}$ is recalculated for the new mixture using the equilibrium vapor phase composition $y_i^{(k-1)}$ from the previous step.

6 Results

6.1 Validation. The described algorithms were verified on a variety of mixtures and experimental data [37–42]. In general, SRK EOS shows good correlations for hydrocarbons. Significant discrepancies can be observed for highly polar substances

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Table 4 Propane (1)-cyclohexane (2); experimental [37] and calculated equilibrium data; $k_{12}=0.016$

P (atm)	T_{exp} (K)	T_{calc} (K)	T_{NIST} (K)	$x_{1\ exp}$	$y_{1\ exp}$	$y_{1\ calc}$	$y_{1\ NIST}$
2.52	313	312.4	281	0.152	0.906	0.8286	0.96
5.47	313	311.2	296	0.378	0.962	0.9427	0.97
8.44	313	311.9	306	0.615	0.9813	0.9776	0.98
11.45	313	312.8	310.5	0.847	0.9922	0.9934	0.99
13.01	313	312.1	313	0.97	0.9983	0.9989	0.999
9.18	393	389.5	354	0.138	0.68	0.7016	0.805
12.44	393	389.1	357	0.208	0.757	0.7928	0.84
16.09	393	388.8	360.5	0.287	0.812	0.8538	0.886
21.42	393			0.395	0.855		
26.05	393	390.2	373	0.491	0.881	0.932	0.88
38.29	393	391.6	382	0.731	0.908	0.975	0.91
17.76	473	470.2	450	0.054	0.232	0.349	0.3
28.62	473	470.4	433	0.175	0.49	0.671	0.58
46.98	473	473.6	438	0.372	0.619	0.852	—
58.23	473	475.7	442	0.515	0.608	0.913	—

(methanol-water mixture) and hydrocarbons with a big size difference (methane-decane and methane-propyl-benzene). Nevertheless, the calculation of the ethane-methylcyclohexane mixture provides satisfactory results.

Table 4 shows the correlation of calculations with experimental data [37] for propane-cyclohexane. The calculations performed also for propane-toluene, propane-m-xylene mixtures [37] demonstrate the same conformity with the experimental data. For the calculations, pressure and liquid phase compositions were preset, equilibrium temperature and vapor phase compositions were predicted.

Mixing rules' parameters k_{ij} are listed in Table 4 along with the results obtained using the REFPROP® code [43] provided by the National Institute of Standards and Technology (NIST). This code uses the Span–Wagner modification [25] of the Benedict–Webb–Rubin (BWR) EOS (4).

REFPROP® [43] has an extensive but nonexhaustive list of species and can have problems with convergence near critical points, which cannot be evaluated with this tool. Model validation for the nine-species mixture [40] $H_2S—CO_2—N_2—CH_4—C_2H_6—C_3H_8$-n-$C_4H_{10}$-iso-$C_4H_{10}—C_5H_{12}$ is presented in Fig. 6. The calculation was made both by using the developed code and REFPROP® [43]. A comparison of the calculated evaporation line with the experimental data for three component mixtures $H_2S—C_6H_{14}—C_{15}H_{32}$ at constant temperature 424.5 K [39] is shown in Fig. 7.

The results of the model validation on experimental data for critical point can be seen in Tables 5 and 6 and Fig. 8.

Figure 9 shows the comparison of calculated and experimental distillation curves. Due to the time delay in the determination of the experimental data, there is a horizontal offset on the volume fraction axis, so the experimental points, except for the point corresponding to the initial boiling temperature, are subject to a sev-

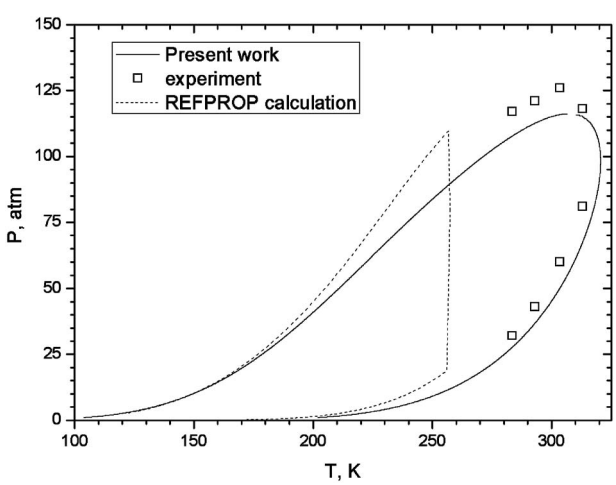

Fig. 6 Calculated and experimental [40] phase diagram for the $H_2S—CO_2—N_2—CH_4—C_2H_6—C_3H_8$-n-$C_4H_{10}$-iso-$C_4H_{10}—C_5H_{12}$ mixture

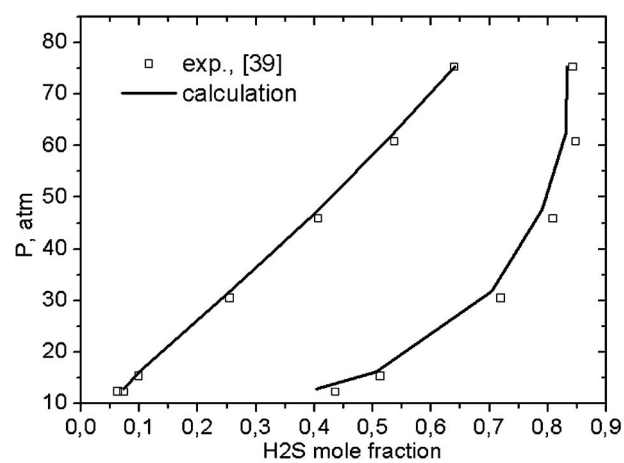

Fig. 7 Equilibrium pressure versus H_2S mole fraction in $H_2S—C_6H_{14}—C_{15}H_{32}$ mixture [39]; $k_{12}=0.07$, $k_{13}=k_{23}=0$

Table 5 Compositions of seven hydrocarbon mixtures, as well as the experimental [41] and calculated critical temperatures and pressures

iso-C_5	0.1989
n-C_5	0.1963
n-C_6	0.1483
n-C_7	0.1344
n-C_8	0.1213
n-C_9	0.1137
n-C_{10}	0.1142
T_{crit} exp., K	543.37
T_{crit} calc., K	543.3
P_{crit} exp., atm	35.04
P_{crit} calc., atm	32.68

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Table 6 Compositions of five hydrocarbon mixtures, as well as the experimental [41] and calculated critical temperatures and pressures

n-C_5	0.2465
n-C_6	0.2176
n-C_7	0.1925
n-C_8	0.1779
n-C_9	0.1656
T_{crit} exp., K	541.26
T_{crit} calc., K	541.14
P_{crit} exp., atm	30.6
P_{crit} calc., atm	30.49

eral percent adjustment, depending on experimental apparatus and operating conditions. The calculations for $C_{10}H_{22}$—$C_{14}H_{30}$ mixtures as well as those suggested in Ref. [17], S-8 fuel surrogate show a very good correlation with the experiments.

The presented results of the model validation allow us to conclude that the selected and well tested algorithms describe the thermodynamic phase equilibrium of multicomponent mixtures reasonably well and can be successfully applied to the evaluation of evaporation characteristics of surrogate blends.

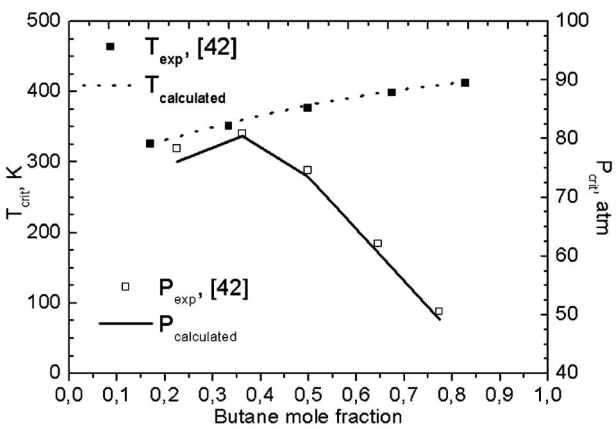

Fig. 8 Calculated and experimental [42] dependencies of critical pressure and temperature for the n-butane-carbon dioxide system. Mixing rules' parameter k_{ij}=0.06.

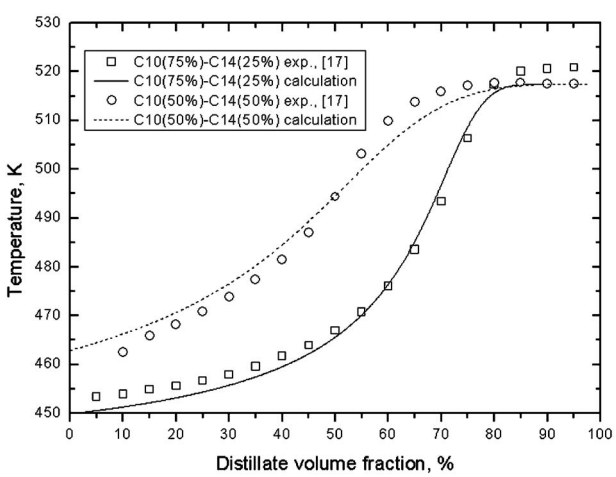

Fig. 9 Experimental and calculated distillation curves for 75/25 and 50/50 mixtures of $C_{10}H_{22}/C_{14}H_{30}$ at P=83 kPa. Experimental data and surrogate mixture are from Ref. [17].

Table 7 Compositions of tested mixtures

Lindstedt and Maurice [7]	n-decane 89% and toluene 11%
Mawid et al. [12]	MCH 5%, toluene 20%, n-decane 25%, iso-octane 5%, n-dodecane 25%, and n-tetradecane 20%
Dagaut and Cathonnet [2]	n-decane 74%, n-propylbenzene 15%, and n-propylcyclohexane 11%
Violi et al., mix No. 2 [10]	Xylenes 8.5%, tetralin 8%, n-octane 3.5%, decalin 35%, n-dodecane 40%, and n-hexadecane 5%
Schulz [45]	Iso-octane 5.7%, MCH 5.1%, m-xylene 4.5%, cyclooctane 4.7%, decane 16.2%, butyl benzene 4.6%, 1,2,4,5-tetramethyl benzene 4.4%, tetralin 4.1%, dodecane 21%, and 1-methyl naphthalene 3.95%

6.2 Two-Phase Equilibrium Calculation. The tools developed for two-phase equilibrium, critical points, and distillation curve calculations have been applied first to examine the predictive capabilities of the surrogate models, presented in literature, Table 7, to reflect the phase equilibrium. Only in Refs. [10,45] did authors use the distillation curve among the other properties for the determination of the chemical composition of the surrogate model. Other proposed fuel models were tested only for chemical property modeling. All suggested surrogates show good results in concentration profiles and ignition delay reproductions [2,7,10,44,45], but they do not describe the two-phase diagram. In Fig. 10 the comparison of phase diagrams calculated for mixtures from Refs. [2,7,10,44,45] and experimental data cited in Ref. [8] are shown. While the critical pressure is described rather satisfactorily, the discrepancies in critical temperatures are remarkably violent. We can say that it is possible to reproduce well the heat release of a fuel with a model, which does not reflect the evaporation properties of the fuel at all. For example, in Figs. 11 and 12 we show the distillation curve and two-phase diagram for gasoline RD 387 [46] and its reference model proposed in Ref. [46] as a blend of 63% n-C_7H_{16}+17% iso-C_8H_{18}+20% C_7H_8. Obviously, the proposed mixture does not describe the evaporation characteristic of the fuel, but the ignition delay times modeled with these surrogates show good conformity with the experimental data [47]. The explanation for that lies in the different nature of modeled processes and in the difference between the methods applied in the chemical and physical modeling.

The chemical models of hydrocarbon combustion have hierarchical structure. Molecules and radicals have the similar func-

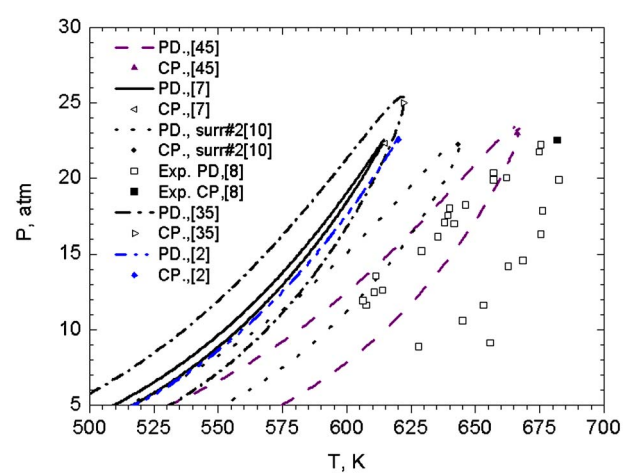

Fig. 10 Calculated phase diagram for suggested in Refs. [2,7,10,44,45] fuel surrogates compared with experimental data cited in Ref. [8] for jet-A. CP means critical point and PD means phase diagram.

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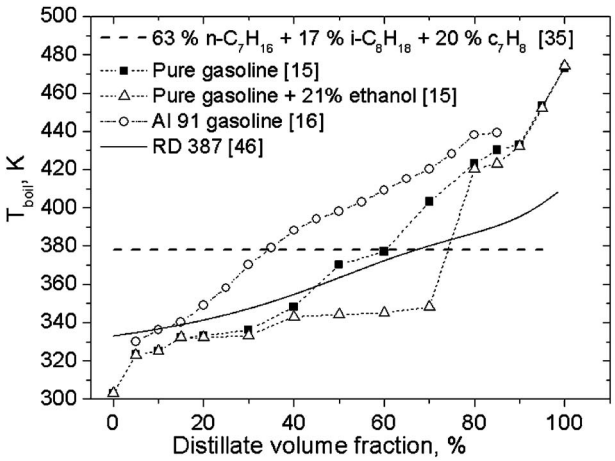

Fig. 11 Experimental distillation curves for gasoline [15,16] and calculated boiling-point curve for RD 387 and its reference fuel [46]

tional groups and consequently similar reaction rates for smaller and higher hydrocarbons, which determine the rate of combustion. While the behavior of functional groups is the key for chemical time scales, physical properties depend more upon intermolecular forces and molecular weight and therefore hydrocarbons with similar functional groups—may differ strongly in physical properties. Also, the temperature of the processes, for which the blends are being analyzed, differs.

Chemical combustion modeling deals mostly with $T > 800$ K, while the temperature of the two-phase diagram is less than 700 K. It means the reference fuel model applied for the computational fluid dynamics (CFD) modeling of the practical combustion chamber must accumulate different capabilities of fuel, which poorly correlate with each other. We can conclude that by the reference model construction, the evaporation facilities of the reference fuel must be tested in the first place. Using the developed tools we optimized the reference mixture with 12 components proposed in Ref. [45], Table 7, to reproduce the phase-equilibrium phenomena. The surrogate blend [45] was selected for optimization because its phase diagram is at its closest to the experimental data, Fig. 10. During the optimization we reduced the number of components in the initial formula of a surrogate [45] to five components, which represent paraffins, naphthenes, and aromatics in the mixture, Table 8. This mixture represents the evaporation properties of kerosene: Figs. 13 and 14 show a good correlation

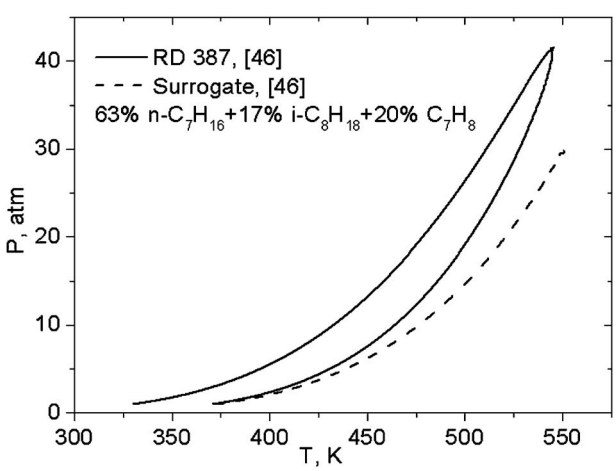

Fig. 12 Calculated two-phase diagram for RD 387 and its reference fuel [46]

Table 8 Composition of a proposed blend for kerosene combustion and its basic properties

10% C_9H_{18} propylcyclohexane	Combustion enthalpy $\Delta_c H_f$	45 MJ/kg
13% iso-C_8H_{18} iso-octane	Formation enthalpy ΔH_f	−160 kJ/mol
20% $C_{12}H_{26}$ dodecane	Molar weight	145 g/mol
25% $C_{11}H_{10}$ 1-methylnaphtalene	Approximate formula	$C_{11}H_{19}$
32% $C_{16}H_{34}$ hexadecane	Sooting tendency index	27.8

with the experimental data and calculated distillation curve and phase diagram. Other essential physical properties presented in Table 8 are in accordance with those of real kerosene. The chemical capabilities of this mixture will be developed later.

7 Conclusion

This paper establishes the principles of the model design of surrogate blends. Next, physical properties of the reference fuel were taken into consideration: combustion enthalpy, formation en-

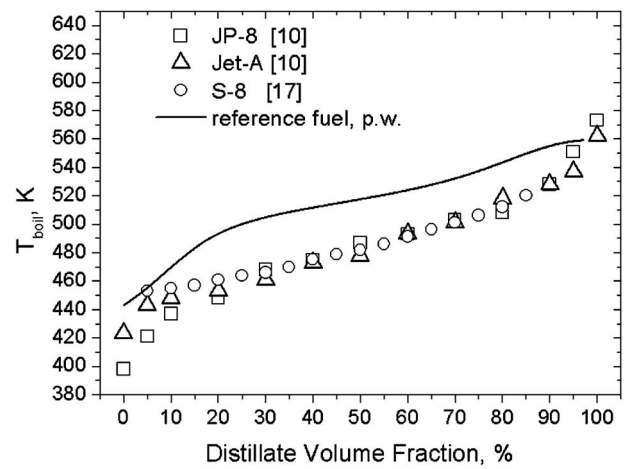

Fig. 13 Experimental distillation curves for kerosene samples from Refs. [10,17] and calculated fuel boiling-point curve for a proposed fuel surrogate

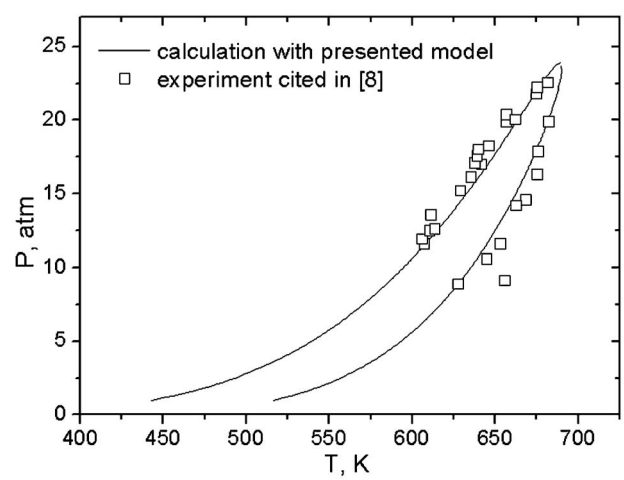

Fig. 14 Calculated phase diagram for a proposed fuel surrogate, Table 6, compared with experimental data cited in Ref. [8] for jet-A

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thalpy, molar weight, approximate formula (C/H ratio), sooting tendency index, critical point, two-phase diagram, and distillation curve. Formulation strategy, experimental data, and numerical tools for blending pure hydrocarbons to reproduce a fuel surrogate distillation curve and phase-equilibrium diagram were described in detail. The efficiency of some surrogate blends, represented in literature, in reflecting the evaporation characteristics was analyzed. Based on the analysis, the chemical capabilities of surrogate models are not linked to their abilities to reflect the phase-equilibrium properties of the fuel and as a consequence good kinetic models cannot represent the thermodynamical behavior of the real system. In general, blending of pure hydrocarbons must begin with the phase-equilibrium and distillation curve calculations.

The selected surrogate mixture closely matches the boiling-point curve and two-phase diagram for jet-A. In the future, viscosity coefficients and the kinetic characteristics of the proposed blend will be investigated.

Acknowledgment

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Nomenclature

A = Helmholtz energy
a = cubic EOS parameter
b = cubic EOS parameter
C = cubic form
f = fugacity
F = functions
H = specific enthalpy
i,j = indices
k = mixing rules' parameter
K = equilibrium constant
M = total number of moles in the liquid phase
n = variation in concentration
N = total number of substances
P = pressure, atm
$\boldsymbol{Q}$ = matrix
$\boldsymbol{Q}_{ij}$ = element of matrix $\boldsymbol{Q}$
R = universal gas constant
t = time
T = temperature
$\boldsymbol{V}$ = vector of variables
V = molar flow rate of evaporation
x = molar fraction in liquid
y = molar fraction in vapor
Z,z = compressibility

Greek Symbols

α,β = temporary variables
δ = Kronecker symbol
ε = cubic EOS parameter
φ = fugacity coefficient
γ = cubic EOS parameter
κ = temporary variable
ρ = density
τ = inverse reduced temperature T_c/T
ν = molar volume
ω = acentric factor
ξ = reduced density ρ/ρ_c

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